

Computer Network, Data Communication and Network Security

Hot Topics

- | | |
|---------------------------|--------------------------|
| ✓ NAT | ✓ WIMAX |
| ✓ PAT | ✓ Cellular Network |
| ✓ IP address related math | ✓ Bluetooth |
| ✓ WIFI | ✓ TCP UDP |
| | ✓ Public and Private Key |

1. Differentiate between Computer Network and Distributed system.

In a distributed system, a collection of independent computers appears to its users as a single coherent system. Usually, it has a single model or paradigm that it presents to the users. Often a layer of software on top of the operating system, called middleware, is responsible for implementing this model. A well-known example of a distributed system is the World Wide Web. It runs on top of the Internet and presents a model in which everything looks like a document (Web page). In a computer network, this coherence, model, and software are absent. If the machines have different hardware and different operating systems, that is fully visible.

2. Differentiate between connection oriented and connectionless services. [BUET]

Connection Oriented:

Connection-oriented service is modeled after the telephone system. To talk to someone, you pick up the phone, dial the number, talk, and then hang up. Similarly, to use a connection-oriented network service, the service user first establishes a connection, uses the connection, and then releases the connection. The essential aspect of a connection is that it acts like a tube: the sender pushes objects (bits) in at one end, and the receiver takes them out at the other end. A typical situation in which a reliable connection-oriented service is appropriate is file transfer. The owner of the file wants to be sure that all the bits arrive correctly and in the same order they were sent.

Connectionless:

connection-oriented service, connectionless service is modeled after the postal system. Each message (letter) carries the full destination address, and each one is routed through the intermediate nodes inside the system independent of all the subsequent messages. There are different names for messages in different contexts; a packet is a message at the network layer. When the intermediate nodes receive a message in full before sending it on to the next node, this is called store-and-forward switching.

3. Write short Note on Twisted Pair.

One of the oldest and still most common transmission media is twisted pair. A twisted pair consists of two insulated copper wires, typically about 1 mm thick. The wires are twisted together in a helical form, just like a DNA molecule. Twisting is done because two parallel wires constitute a fine antenna. When the wires are twisted, the waves from different twists cancel out, so the wire radiates less effectively. A signal is usually carried as the difference in voltage between the two wires in the pair. This provides better immunity to external noise because the noise tends to affect both wires the same, leaving the differential unchanged. The most common application of the twisted pair is the telephone system.

4. Describe OSI model with appropriate diagram.

The model is called the ISO OSI (Open Systems Interconnection) Reference Model because it deals with connecting open systems—that is, systems that are open for communication with other systems.

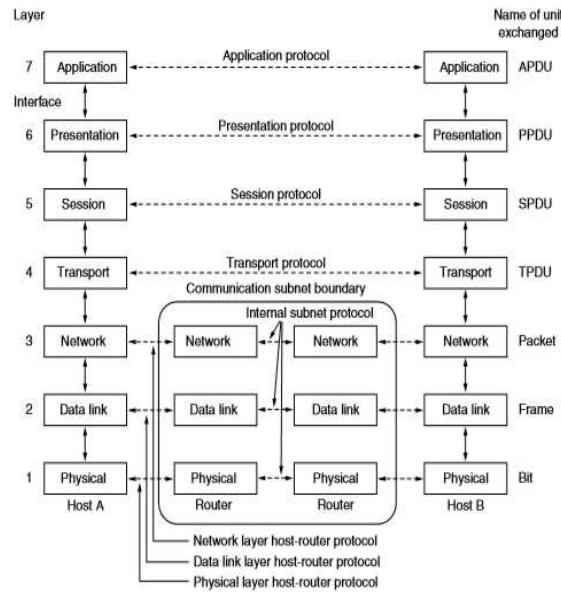


Figure 1-20. The OSI reference model.

Physical Layer:

- Does the job of carrying bit streams(bit by bit) over a physical medium.
- Keeps track of signals.
- Defines the duration of a bit.
- Bit encoding
- Protocols/connection: Ethernet, HUB, Repeater

Data Link Layer:

- Transforms bits into frames.
- Provides node to node data transfer.
- Defines the error detection and correction schemes.
- Protocols/connection: ATM, SWITCH, BRIDGE

Network Layer:

- Does the job of data transmission from source to destination.
- Translates logical network address in physical machine address.
- Routing and subnetting.
- Reliable message delivery.
- Protocols/connection: IP, IPX, Routing

Transport layer:

- End to end communication.
- Provides logical communication between application process running and hosts.
- Management of error correction.
- Transmission reliability.
- Protocols/connection: TCP,UDP

Session Layer:

- Does the job of control over network.
- Establishes, maintains and terminates connections.
- Provides the mechanism for opening, closing and managing a session.
- Protocols/connection: Apple talk

Presentation Layer:

- Concerns with syntax and semantics of the information.
- Translates data into suitable format.
- Data encryption
- Protocols/connection: JPEG, GIF

Application Layer:

- Provides user interface.
- Provides mail services.
- Protocols/connection: HTTP, FTP, SMTP, DNS etc

5. Differentiate TCP/IP and OSI model.

TCP/IP	OSI
Implementation of OSI model	Reference model
Model around which Internet is developed	This is a theoretical model
Has only 4 layers	Has 7 layers
Considered more reliable	Considered a reference tool
Protocols are not strictly defined	Stricter boundaries for the protocols
Horizontal approach	Vertical approach
Combines the session and presentation layer in the application layer	Has separate session and presentation layer
Protocols were developed first and then the model was developed	Model was developed before the development of protocols
Supports only connectionless communication in the network layer	Supports connectionless and connection-oriented communication in the network layer

6. Describe TCP/IP model with appropriate diagram.

TCP/IP is a protocol suit or framework for Internet. TCP is used for reliable connection and IP is used for addressing hosts. It has four layers. They are:

The Link Layer

The lowest layer in the model, the link layer describes what links such as serial lines and classic Ethernet must do to meet the needs of this connectionless internet layer. It is not really a layer at all, in the normal sense of the term, but rather an interface between hosts and transmission links.

The Internet Layer

The internet layer is the linchpin that holds the whole architecture together. Its job is to permit hosts to inject packets into any network and have them travel independently to the destination (potentially on a different network). They may even arrive in a completely different order than they were sent, in which case it is the job of higher layers to rearrange them, if in-order delivery is desired. The internet layer defines an official packet format and protocol called IP (Internet Protocol), plus a companion protocol called ICMP (Internet Control Message Protocol) that helps it function. The job of the internet layer is to deliver IP packets where they are supposed to go. Packet routing is clearly a major issue here, as is congestion (though IP has not proven effective at avoiding congestion).

The Transport Layer

The layer above the internet layer in the TCP/IP model is now usually called the transport layer. It is designed to allow peer entities on the source and destination hosts to carry on a conversation, just as in the OSI transport layer. Two end-to-end transport protocols have been defined here. The first one, TCP (Transmission Control Protocol), is a reliable connection-oriented protocol that allows a byte stream originating on one machine to be delivered without error on any other machine in the internet. It segments the incoming byte stream into discrete messages and passes each one on to the internet layer. At the destination, the receiving TCP process reassembles the received messages into the output stream. TCP also handles flow control to make sure a fast sender cannot swamp a slow receiver with more messages than it can handle. The second protocol in this layer, UDP (User Datagram Protocol), is an unreliable, connectionless protocol for applications that do not want TCP's sequencing or flow control and wish to provide their own.

The Application Layer

The TCP/IP model does not have session or presentation layers. No need for them was perceived. On top of the transport layer is the application layer. It contains all the higher-level protocols. The early ones included virtual terminal (TELNET), file transfer (FTP), and electronic mail (SMTP).

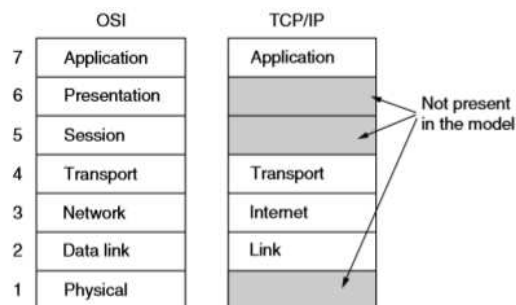


Figure 1-21. The TCP/IP reference model.

7. What do you know about ARPANET?

A packet-switched network connecting a number of LANs, called Arpanet developed by ARPA (Advanced Research Projects Agency). Originally, Arpanet was strictly military and defense-oriented. Arpanet was converted to use the new standard TCP/IP protocol set (1980)

ARPANET is the grandparent of all wide area computer networks. The ARPANET was a research network sponsored by the DoD (U.S. Department of Defense). It eventually connected hundreds of universities and government installations, using leased telephone lines. When satellite and radio networks were added later, the existing protocols had trouble interworking with them, so a new reference architecture was needed. Thus, from nearly the beginning, the ability to connect multiple networks in a seamless way was one of the major design goals. This architecture later became known as the TCP/IP Reference Model, after its two primary protocols.

The Defense Communication Agency (DCA) split Arpanet into two networks (1983):

Arpanet: To be used for internetworking research projects

Milnet: To be used strictly for military purposes

8. What is RFID?

RFID: RFID (radio frequency identification) is a form of wireless communication that incorporates the use of electromagnetic or electrostatic coupling in the radio frequency portion of the electromagnetic spectrum to uniquely identify an object, animal or person. Use cases for RFID technology include healthcare, manufacturing, inventory management, shipping, retail sales and home use.

Every RFID system consists of three components: a scanning antenna, a transceiver and a transponder. When the scanning antenna and transceiver are combined, they are referred to as an RFID reader or interrogator. The RFID reader is a network-connected device that can be portable or permanently attached. It uses radio frequency waves to transmit signals that activate the tag. Once activated, the tag sends a wave back to the antenna, where it is translated into data.

The transponder is located in the RFID tag itself. The read range for RFID tags varies based on factors including the type of tag, type of reader, RFID frequency and interference in the surrounding environment or from other RFID tags and readers. Generally speaking, tags that have a stronger power source also have a longer read range.



Figure 4: RFID System

9. Write Short Notes on Coaxial Cable.

Another common transmission medium is the coaxial cable (known to its many friends as just “coax” and pronounced “co-ax”). It has better shielding and greater bandwidth than unshielded twisted pairs, so it can span longer distances at higher speeds. Two kinds of coaxial cable are widely used. One kind, 50-ohm cable, is commonly used when it is intended for digital transmission from the start. The other kind, 75-ohm cable, is commonly used for analog transmission and cable television. This distinction is based on historical, rather than technical, factors (e.g., early dipole antennas had an impedance of 300 ohms, and it was easy to use existing 4:1 impedance-matching transformers). Starting in the mid1990s, cable TV operators began to provide Internet access over cable, which has made 75-ohm cable more important for data communication. A coaxial cable consists of a stiff copper wire as the core, surrounded by an insulating material. The insulator is encased by a cylindrical conductor, often as a closely woven braided mesh. The outer conductor is covered in a protective plastic sheath.

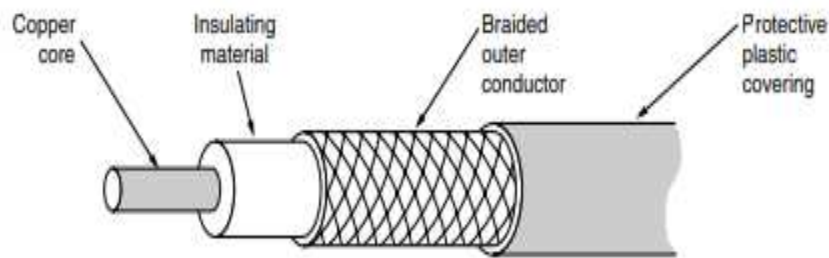


Figure 2-4. A coaxial cable.

10. Describe fibre optics cables.

A fiber optic cable is a network cable that contains strands of glass fibers inside an insulated casing. They're designed for long-distance, high-performance data networking, and telecommunications. Compared to wired cables, fiber optic cables provide higher bandwidth and transmit data over longer distances. Fiber optic cables support much of the world's internet, cable television, and telephone systems.

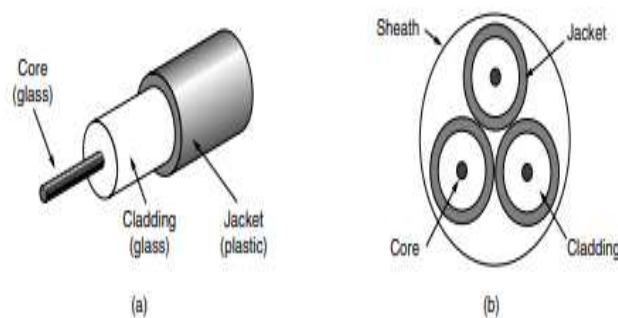


Figure 2-8. (a) Side view of a single fiber. (b) End view of a sheath with three fibers.

11. Describe VSATs.

A recent development in the communication satellite world is the development of low-cost MicroStation's, sometimes called VSATs (Very Small Aperture Terminals). These tiny terminals have 1-meter or smaller antennas (versus 10 m for a standard GEO antenna) and can put out about 1 watt of power. The uplink is generally good for up to 1 Mbps, but the downlink is often up to several megabits/sec. Direct broadcast satellite television uses this technology for one-way transmission. In many VSAT systems, the MicroStation's do not have enough power to communicate directly with one another (via the satellite, of course). Instead, a special ground station, the hub, with a large, high-gain antenna is needed to relay traffic between VSATs.

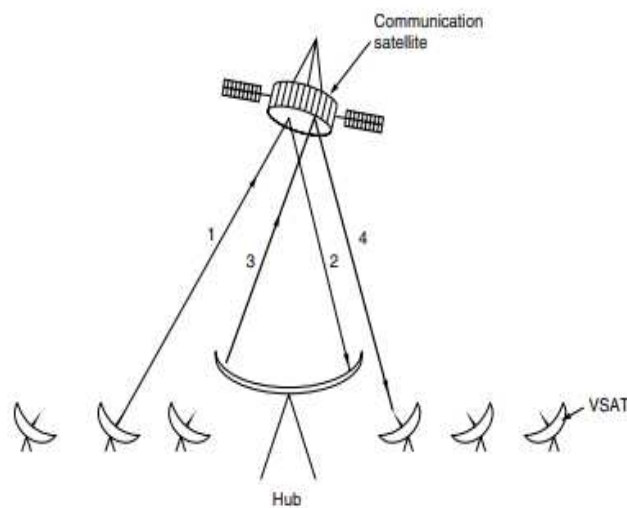


Figure 2-17. VSATs using a hub.

12. Describe SMTP

SMTP is part of the application layer of the TCP/IP protocol. Using a process called “store and forward,” SMTP moves your email on and across networks. It works closely with something called the Mail Transfer Agent (MTA) to send your communication to the right computer and email inbox.

SMTP spells out and directs how your email moves from your computer's MTA to an MTA on another computer, and even several computers. Using that “store and forward” feature mentioned before, the message can move in steps from your computer to its destination. At each step, Simple Mail Transfer Protocol is doing its job. Lucky for us, this all takes place behind the scenes, and we don't need to understand or operate SMTP.

13. Describe Circuit switching and packet switching with diagram.

Circuit Switching: Circuit switching is defined as the method of switching which is used for establishing a dedicated communication path between the sender and the receiver. The link which is established between the sender and the receiver is in the physical form. Analog telephone network is a well-known example of circuit switching.

Advantages of Circuit Switching

- The bandwidth used is fixed.
- The quality of communication is increased as a dedicated communication channel is used.
- The rate at which the data is transmitted is fixed.
- While switching, no time is wasted in waiting.
- It is preferred when the communication is long and continuous.

Disadvantages of Circuit Switching

- Since dedicated channels are used, the bandwidth required is more.
- The utilization of resources is not full.
- Since a dedicated channel has been used, the transmission of other data becomes impossible.
- The time taken by the two stations for the establishment of the physical link is too long.
- Circuit switching is expensive because every connection uses a dedicated path establishment.
- The link between the sender and the receiver will be maintained until and unless the user terminates the link. This will also continue if there is no transfer of data taking place.

Packet Switching

Packet switching is defined as the connectionless network where the messages are divided and grouped together and, this is known as a packet. Each packet is routed from the source to the destination as individual packets. The actual data in these packets are carried by the payload. When the packet arrives at the destination, it is the responsibility of the destination to put these packets in the right order.

Advantages of Packet Switching

- There is no delay in the delivery of the packets as they are sent to the destination as soon as they are available.
- There is no requirement for massive storage space as the information is passed on to the destination as soon as they are received.
- Failure in the links does not stop the delivery of the data as these packets can be routed from other paths too.
- Multiple users can use the same channel while transferring their packets.
- The usage of bandwidth is better in case of packet switching as multiple sources can transfer packets from the same source link.

Disadvantages of Packet Switching

- Installation costs of packet switching are expensive.
- The delivery of these packets becomes easy when complicated protocols are used.
- High-quality voice calls cannot use packet switching as there is a lot of delay in this type of communication.
- Connectivity issues may lead to loss of information and delay in the delivery of the information.

14. What is MAC? Write down the importance of MAC.

MAC: MAC stands for Media Access Control. It is a unique identifier for network interfaces. The MAC address is usually assigned by the manufacturer of a Network Interface Controller (NIC), and it is stored in the hardware. The NIC is a computer circuit card that allows a computer to connect to a network. During network communication the Address Resolution Protocol (ARP) for the Internet Protocol Version 4 (IPv4) or the Neighbor Discovery Protocol (NDP) for IPv6 translates the IP into a NIC. The MAC address is formed in accordance to the rules of the three numbering name spaces, which are managed by the Institute of Electrical and Electronic Engineers (IEEE). The format is six sets of two digits or characters, separated by hyphens. An example of a MAC address is 30-65-EC-6F-C4-58.

Importance of MAC address:

One of the applications of MAC addresses is in the filtering process on wireless networks. In order to prevent strangers from accessing a network, the router is set to accept only specific MAC addresses. In this manner, if the IP address changes, as for example in the case of dynamic IP addresses, the MAC address can still identify the device. Filtering can be used to track network users, and to limit their access. It can also have other uses, such as identifying when a stolen device connects to the network. For these reasons, many companies and institutions require the MAC addresses of their members' devices. Therefore, it is important for device owners not to reveal their MAC addresses to anyone, except to authorized personnel.

15. What is DNS? Describe briefly how DNS Work.

DNS: The Domain Name System (DNS) is the phonebook of the Internet. Humans access information online through domain names, like nytimes.com or espn.com. Web browsers interact through Internet Protocol (IP) addresses. DNS translates domain names to IP addresses so browsers can load Internet resources. Each device connected to the Internet has a unique IP address which other machines use to find the device. DNS servers eliminate the need for humans to memorize IP addresses such as 192.168.1.1 (in IPv4), or more complex newer alphanumeric IP addresses such as 2400:cb00:2048:1::c629:d7a2 (in IPv6).

The process of DNS resolution involves converting a hostname (such as www.example.com) into a computer-friendly IP address (such as 192.168.1.1). An IP address is given to each device on the Internet, and that address is necessary to find the appropriate Internet device - like a street address is used to find a particular home. When a user wants to load a webpage, a translation must occur between what a user types into their web browser (example.com) and the machine-friendly address necessary to locate the example.com webpage.

In order to understand the process behind the DNS resolution, it's important to learn about the different hardware components a DNS query must pass between. For the web browser, the DNS lookup occurs “behind the scenes” and requires no interaction from the user's computer apart from the initial request.

16. Mansion some error correcting codes.

four different error-correcting codes are: Hamming codes, Binary convolutional codes, Reed-Solomon codes, Low-Density Parity Check codes.

17. Describe Bluetooth Architecture with figure. [BUET]

The IEEE 802.15.1 standard is the basis for the Bluetooth wireless communication technology. Bluetooth is an open wireless technology standard for transmitting fixed and mobile electronic device data over short distances. Bluetooth communicates with a variety of electronic devices and creates personal networks operating within the unlicensed 2.4 GHz band. Operating range is based on device class. A variety of digital devices use Bluetooth, including MP3 players, mobile and peripheral devices and personal computers. The basic unit of a Bluetooth system is a piconet, which consists of a master node and up to seven active slave nodes within a distance of 10 meters. Multiple piconets can exist in the same (large) room and can even be connected via a bridge node that takes part in multiple piconets. An interconnected collection of piconets is called a scatternet.

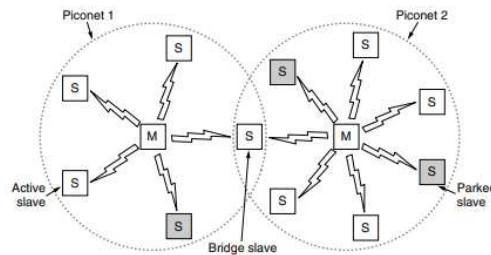


Figure 4-34. Two piconets can be connected to form a scatternet.

18. Define cyclic redundancy check (CRC).

The cyclic redundancy check (CRC) is a technique used to detect errors in digital data. As a type of checksum, the CRC produces a fixed-length data set based on the build of a file or larger data set. In terms of its use, CRC is a hash function that detects accidental changes to raw computer data commonly used in digital telecommunications networks and storage devices such as hard disk drives. In the cyclic redundancy check, a fixed number of check bits, often called a checksum, are appended to the message that needs to be transmitted. The data receivers receive the data, and inspect the check bits for any errors.

19. What is hamming code?

Hamming code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is moved or stored from the sender to the receiver. Redundant bits are extra binary bits that are generated and added to the information-carrying bits of data transfer to ensure that no bits were lost during the data transfer.

The number of redundant bits can be calculated using the following formula:

$$2^r \geq m + r + 1$$

where, r = redundant bit, m = data bit

20. What is Firewall?

A firewall is a division between a private network and an outer network, often the internet, that manages traffic passing between the two networks. It's implemented through either hardware or software. Firewalls allow, limit, and block network traffic based on preconfigured rules in the hardware or software, analyzing data packets that request entry to the network. In addition to limiting access to computers and networks, a firewall is also useful for allowing remote access to a private network through secure authentication certificates and logins.

21. What is Repeater?

Repeaters are network devices operating at physical layer of the OSI model that amplify or regenerate an incoming signal before retransmitting it. They are incorporated in networks to expand its coverage area. They are also known as signal boosters.

When an electrical signal is transmitted via a channel, it gets attenuated depending upon the nature of the channel or the technology. This poses a limitation upon the length of the LAN or coverage area of cellular networks. This problem is alleviated by installing repeaters at certain intervals. Repeaters amplifies the attenuated signal and then retransmits it. Digital repeaters can even reconstruct signals distorted by transmission loss. So, repeaters are popularly incorporated to connect between two LANs thus forming a large single LAN.

22. What is TELNET?

TELNET stands for Terminal Network. It is a type of protocol that enables one computer to connect to local computer. It is used as a standard TCP/IP protocol for virtual terminal service which is given by ISO. Computer which starts connection known as the local computer. Computer which is being connected to i.e. which accepts the connection known as remote computer. When the connection is established between local and remote computer. During telnet operation whatever that is being performed on the remote computer will be displayed by local computer. Telnet operates on client/server principle. Local computer uses telnet client program and the remote computers uses telnet server program.

23. What is MANET?

MANET Stands for "Mobile Ad Hoc Network." A MANET is a type of ad hoc network that can change locations and configure itself on the fly. Because MANETS are mobile, they use wireless connections to connect to various networks. This can be a standard Wi-Fi connection, or another medium, such as a cellular or satellite transmission

24. What do you understand by Multiplexing and Demultiplexing?

Whenever an entity accepts items from more than one source, this is referred to as multiplexing (many to one). Whenever an entity delivers items to more than one source, this is referred to as demultiplexing (one to many). The transport layer at the source performs multiplexing The transport layer at the destination performs demultiplexing.

25. What do you understand by unicast, multicast and broadcast?

Unicast Address: Each host or each interface of a router is assigned a unicast address. Unicasting means one-to-one communication. A frame with a unicast address destination is destined only for one entity in the link.

Multicast Address: Link-layer protocols define multicast addresses. Multicasting means one-to-many communication but not all.

Broadcast Address: Link-layer protocols define a broadcast address. Broadcasting means one-to-all communication. A frame with a destination broadcast address is sent to all entities in the link.

26. What do you know about HUB, Repeater, Bridge, Switch, Router and gateway?

HUBS

Several networks need a central location to connect media segments together. These central locations are called as hubs. The hub organizes the cables and transmits incoming signals to the other media segments.

The three types of hubs are:

- i) **Passive hub:** It is a connector, which connects wires coming from the different branches. By using passive hub, each computer can receive the signal which is sent from all other computers connected in the hub.
- ii) **Active Hub:** It is a multiport repeater, which can regenerate the signal. It is used to create connections between two or more stations in a physical star topology.
- iii) **Intelligent Hub:** Intelligent hub contains a program of network management and intelligent path selection.

REPEATERS

A repeater receives the signal and it regenerates the signal in original bit pattern before the signal gets too weak or corrupted. It is used to extend the physical distance of LAN. Repeater works on physical layer. A repeater has no filtering capability. A repeater is implemented in computer networks to expand the coverage area of the network, repropagate a weak or broken signal and or service remote nodes. Repeaters amplify the received/input signal to a higher frequency domain so that it is reusable, scalable and available.

BRIDGES

Bridges operate in physical layer as well as data link layer. As a physical layer device, they regenerate the receive signal. As a data link layer, the bridge checks the physical (MAC) address (of the source and the destination) contained in the frame. The bridge has a filtering feature. It can check the destination address of a frame and decides, if the frame should be forwarded or dropped. Bridges are used to connect two or LANs working on the same protocol.

SWITCHES

A switch is a small hardware device which is used to join multiple computers together with one local area network (LAN). A switch is a mechanism that allows us to interconnect links to form a large network. Switch is data link layer device. A switch is a multi-port bridge with a buffer. Switches are used to forward the packets based on MAC addresses. It is operated in full duplex mode. Packet collision is minimum as it directly communicates between source and destination. It does not broadcast the message as it works with limited bandwidth. A switch's primary job is to receive incoming packets on one of its links and to transmit them on some other link. A Switch is used to transfer the data only to the device that has been addressed.

ROUTERS

A router is a three-layer device. It operates in the physical, data-link, and network layers. As a physical-layer device, it regenerates the signal it receives. As a link-layer device, the router checks the physical addresses (source and destination) contained in the packet. As a network-layer device, a router checks the network-layer addresses. A router is a device like a switch that routes data packets based on their IP addresses. A router can connect networks. A router connects the LANs and WANs on the internet. A router is an internetworking device. It connects independent networks to form an internetwork. The key function of the router is to determine the shortest path to the destination. Router has a routing table, which is used to make decision on selecting the route. The routing table is updated dynamically based on which they make decisions on routing the data packets.

GATEWAY

A gateway is a device, which operates in all five layers of the internet or seven layers of OSI model. It is usually a combination of hardware and software. Gateway connects two independent networks. Gateways are generally more complex than switch or router. Gateways basically works as the messenger agents that take data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters A gateway accepts a packet formatted for one protocol and converts it to a packet formatted to another protocol before forwarding it. The gateway must adjust the data rate, size and data format.

27. What are differences between NAT and PAT? [BUET]

Basis	NAT	PAT
Basic	Translates the private local IP address to the public global IP address.	Similar to NAT it also translates the private IP addresses of an internal network to the public IP address with the help of Port numbers.
Relationship	Superset of PAT.	Variant of NAT (form of a Dynamic NAT).
Uses	IPv4 address	IPv4 addresses along with the port number.
Types	Static NAT Dynamic NAT	Static PAT Overloaded PAT

28. What do you mean by flow control, error control, and Congestion control?

Flow Control: Flow Control is the process of managing the rate of data transmission between two nodes to prevent a fast sender from overwhelming a slow receiver. It provides a mechanism for the receiver to control the transmission speed, so that the receiving node is not overwhelmed with data from transmitting node.

Error Control: Error control at the transport layer is responsible for

- Detecting and discarding corrupted packets.
- Keeping track of lost and discarded packets and resending them.
- Recognizing duplicate packets and discarding them.
- Buffering out-of-order packets until the missing packets arrive.

Error Control involves Error Detection and Error Correction.

Congestion Control: Congestion in a network may occur if the load on the network (the number of packets sent to the network) is greater than the capacity of the network (the number of packets a network can handle). Congestion control refers to the mechanisms and techniques that control the congestion and keep the load below the capacity. Congestion Control refers to techniques and mechanisms that can either prevent congestion, before it happens, or remove congestion, after it has happened. Congestion control mechanisms are divided into two categories,

1. Open loop - prevent the congestion before it happens.
2. Closed loop - remove the congestion after it happens.

29. Describe the Transmission rate of wifi, cellular, wimax and Bluetooth [BUET]

<https://www.awirelesslife.com/wireless.html>

Bluetooth

Named after the Danish king, Harold Bluetooth, was the first to emerge, several devices like mobile phones, headsets, keyboards, mice, medical equipment and even cars now come with this feature. Due to its low cost, manufacturers are willing to implement this technology in most devices. It is designed for short range communications with a range of about 10m. As a result, it consumes less power and are suited for very small battery powered devices and portable devices. Problems associated when devices communicate via infrared or cables are removed. Infrared requires a line of sight, Bluetooth only needs to be in reasonable vicinity. As cables are not required, it would be less cumbersome carrying a personal Bluetooth device and space would be less cluttered. As Bluetooth devices automatically communicate with each other, it requires very little from the user. Bluetooth allows for a wireless Personal Area Network (PAN) with its short range. See chart below for a comparison of these technologies. For more technical resources and information relating to Bluetooth see the official Bluetooth site.

WiFi

WiFi or Wireless Fidelity, has a range of about 100m and allows for faster data transfer rate between 10 - 54Mbps. There are three different wireless standards under WiFi, 802.11a, 802.11b and 802.11g. 802.11 being the wireless standard set by The Institute of Electrical and Electronic Engineers (IEEE). WiFi is used to create wireless Local Area Networks (WLAN). The most widely used standard is 802.11b and 802.11g is expected to grow rapidly. These two standards are relatively inexpensive and can be found providing wireless connectivity in airports, railway stations, cafes, bars, restaurants and other public areas. The main difference between the two is the speed. 802.11b has data transfer rate of upto 11Mbps and 802.11g has a rate of upto 54Mbps. 802.11g is a relatively new and has yet to be adopted widely. 802.11a is more expensive and as a result it not available for public access. See chart below for a comparison of these technologies.

WiMAX

WiMAX is Worldwide Interoperability for Microwave Access. The IEEE standard for WiMAX is 802.16 and falls under the category of wireless Metropolitan Area Network (WMAN). WiMAX operates on two frequency bands, 2 - 11GHz and 10 - 66GHz and has a range of about 50km with speeds of upto 80Mbps. This enables smaller wireless LANs to be interconnected by WiMAX creating a large wireless MAN. Networking between cities can be achieved without the need for expensive cabling. It is also able to provide high speed wireless broadband access to users. As it can operate in two frequency bands WiMAX can work by line-of-sight and non-line-of-sight. At the 2 - 11GHz frequency range it works by non-line-of-sight, where a computer inside a building communicates with a tower/antenna outside the building. Short frequency transmissions are not easily disrupted by physical obstructions. Higher frequency transmissions are used for non-line-of-sight service. This enables to towers/antennae to communicate with each other over a greater distance. Due to infrastructure and costs involved it would be more suited to provide the backbone services for ISPs and large corporations providing wireless networking and internet access. See chart below for a comparison of these technologies.

	Bluetooth	WiFi (a)	WiFi (b)	WiFi (g)	WIMAX
Standard	802.15	802.11a	802.11b	802.11g	802.16
Frequency (GHz)	2.45	5	2.4	2.4	2 - 66
Speed (Mbps)	0.72	54	11	54	80
Range	10m	50m	100m	100m	50km
Advantages	Low Cost	Speed	Low Cost	Speed	Speed, Range
Disadvantages	Range	Cost	Speed	Cost, Range	Cost

30. How are keys used in SSL encryption (HTTPS)?

<https://www.cloudflare.com/learning/ssl/what-is-a-cryptographic-key/>

There are two kinds of encryption: symmetric encryption and asymmetric encryption, also known as public key encryption. In symmetric encryption, both sides of a conversation use the same key for turning plaintext into ciphertext and vice versa. In asymmetric or public key encryption, the two sides of the conversation each use a different key. One key is called the public key, and one key is called the private key – thusly named because one of the parties keeps it secret and never shares it with anyone. When plaintext is encrypted with the public key, only the private key can decrypt it, not the public key. It works the other way, too: When plaintext is encrypted with the private key, only the public key can decrypt it.

SSL (or TLS, as it is called today), is an encryption protocol used to keep Internet communications secure, and a website that is served over HTTPS instead of HTTP uses this kind of encryption. In TLS/SSL, a website or web application will have both a public key and a private key. The public key is shared publicly in the website's SSL certificate for anyone to see. The private key is installed on the origin server and never shared.

TLS/SSL communication sessions begin with a TLS handshake, during which the website and the client use the public key and the private key in order to generate new keys, which are called session keys. These session keys are then used by both sides to encrypt their messages back and forth.

Thus, TLS starts with asymmetric encryption (with two keys) and moves to symmetric encryption (with one key). Both sides use the same keys during the communication session, but when they start a new session, they will generate new keys together.

31. What is the role of key in cryptography? Describe Public & Private key encryption algorithm.

[BUET]

In cryptography, a key is a string of characters used within an encryption algorithm for altering data so that it appears random. Like a physical key, it locks (encrypts) data so that only someone with the right key can unlock (decrypt) it. The original data is known as the plaintext, and the data after the key encrypts it is known as the ciphertext.

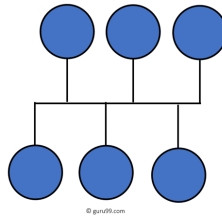
The formula:

plaintext + key = ciphertext

Private Key	Public Key
Private key is faster than public key.	It is slower than private key.
In this, the same key (secret key) and algorithm is used to encrypt and decrypt the message.	In public key cryptography, two keys are used, one key is used for encryption and while the other is used for decryption.
In private key cryptography, the key is kept as a secret.	In public key cryptography, one of the two keys is kept as a secret.
Private key is Symmetrical because there is only one key that is called secret key.	Public key is Asymmetrical because there are two types of key: private and public key.
In this cryptography, sender and receiver need to share the same key.	In this cryptography, sender and receiver does not need to share the same key.
In this cryptography, the key is private.	In this cryptography, public key can be public and private key is private.

Topology: <https://www.guru99.com/type-of-network-topology.html>

Bus Topology



Bus topology uses a single cable which connects all the included nodes. The main cable acts as a spine for the entire network. One of the computers in the network acts as the computer server. When it has two endpoints, it is known as a linear bus topology.

Advantages:

Here are pros/benefits of using a bus topology:

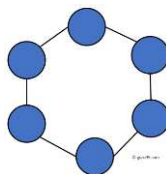
- Cost of the cable is very less as compared to other topology, so it is widely used to build small networks.
- Famous for LAN network because they are inexpensive and easy to install.
- It is widely used when a network installation is small, simple, or temporary.
- It is one of the passive topologies. So computers on the bus only listen for data being sent, that are not responsible for moving the data from one computer to others.

Disadvantages:

Here are the cons/drawbacks of bus topology:

- In case if the common cable fails, then the entire system will crash down.
- When network traffic is heavy, it develops collisions in the network.
- Whenever network traffic is heavy, or nodes are too many, the performance time of the network significantly decreases.
- Cables are always of a limited length.

Ring Topology



In a ring network, every device has exactly two neighboring devices for communication purpose. It is called a ring topology as its formation is like a ring. In this topology, every computer is connected to another computer. Here, the last node is combined with a first one.

This topology uses token to pass the information from one computer to another. In this topology, all the messages travel through a ring in the same direction.

Advantages:

Here are pros/benefits of ring topology:

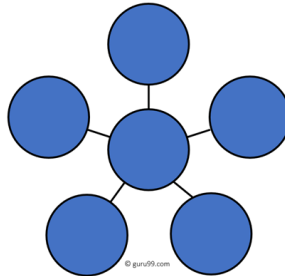
- Easy to install and reconfigure.
- Adding or deleting a device in-ring topology needs you to move only two connections.
- The troubleshooting process is difficult in a ring topology.
- Failure of one computer can disturb the whole network.
- Offers equal access to all the computers of the networks
- Faster error checking and acknowledgment.

Disadvantages:

Here are drawbacks/cons of ring topology:

- Unidirectional traffic.
- Break in a single ring can risk the breaking of the entire network
- Modern days high-speed LANs made this topology less popular.
- In the ring, topology signals are circulating at all times, which develops unwanted power consumption.
- It is very difficult to troubleshoot the ring network.
- Adding or removing the computers can disturb the network activity.

Star Topology



In the star topology, all the computers connect with the help of a hub. This cable is called a central node, and all other nodes are connected using this central node. It is most popular on LAN networks as they are inexpensive and easy to install.

Advantages:

Here are pros/benefits of start topology:

- Easy to troubleshoot, set up, and modify.
- Only those nodes are affected, that has failed. Other nodes still work.
- Fast performance with few nodes and very low network traffic.
- In Star topology, addition, deletion, and moving of the devices are easy.

Disadvantages:

Here are cons/drawbacks of using Star:

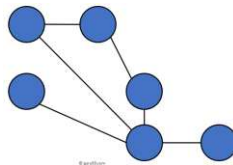
- If the hub or concentrator fails, attached nodes are disabled.
- Cost of installation of star topology is costly.
- Heavy network traffic can sometimes slow the bus considerably.
- Performance depends on the hub's capacity
- A damaged cable or lack of proper termination may bring the network down.

Mesh Topology

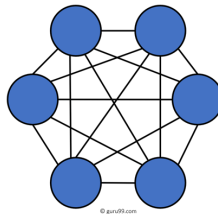
The mesh topology has a unique network design in which each computer on the network connects to every other. It develops a P2P (point-to-point) connection between all the devices of the network. It offers a high level of redundancy, so even if one network cable fails, still data has an alternative path to reach its destination.

Types of Mesh Topology:

- **Partial Mesh Topology:** In this type of topology, most of the devices are connected almost similarly as full topology. The only difference is that few devices are connected with just two or three devices.



- **Full Mesh Topology:** In this topology, every nodes or device are directly connected with each other.



Advantages:

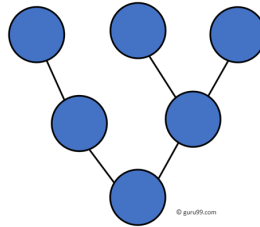
Here, are pros/benefits of Mesh topology

- The network can be expanded without disrupting current users.
- Need extra capable compared with other LAN topologies.
- No traffic problem as nodes has dedicated links.
- Dedicated links help you to eliminate the traffic problem.
- A mesh topology is robust.
- It has multiple links, so if any single route is blocked, then other routes should be used for data communication.
- P2P links make the fault identification isolation process easy.
- It helps you to avoid the chances of network failure by connecting all the systems to a central node.
- Every system has its privacy and security.

Disadvantages:

- Installation is complex because every node is connected to every node.
- It is expensive due to the use of more cables. No proper utilization of systems.
- Complicated implementation.
- It requires more space for dedicated links.
- Because of the amount of cabling and the number of input-outputs, it is expensive to implement.
- It requires a large space to run the cables.

Tree Topology



Tree topologies have a root node, and all other nodes are connected which form a hierarchy. So it is also known as hierarchical topology. This topology integrates various star topologies together in a single bus, so it is known as a Star Bus topology. Tree topology is a very common network which is similar to a bus and star topology.

Advantages:

Here are pros/benefits of tree topology:

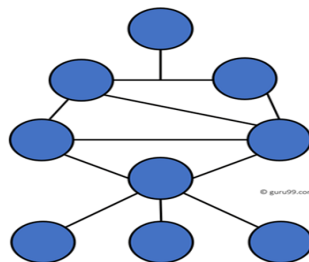
- Failure of one node never affects the rest of the network.
- Node expansion is fast and easy.
- Detection of error is an easy process
- It is easy to manage and maintain

Disadvantages:

Here are cons/drawback of tree topology:

- It is heavily cabled topology
- If more nodes are added, then its maintenance is difficult
- If the hub or concentrator fails, attached nodes are also disabled.

Hybrid Topology



Hybrid topology combines two or more topologies. You can see in the above architecture in such a manner that the resulting network does not exhibit one of the standard topologies.

For example, as you can see in the above image that in an office in one department, Star and P2P topology is used. A hybrid topology is always produced when two different basic network topologies are connected.

Advantages:

Here, are advantages/pros using Hybrid topology:

- Offers the easiest method for error detecting and troubleshooting
- Highly effective and flexible networking topology
- It is scalable so you can increase your network size

Disadvantages:

- The design of hybrid topology is complex
- It is one of the costliest processes

32. How to select a Network Topology?

Here are some important considerations for selecting the best topology to create a network in your organization:

- Bus topology is surely least expensive to install a network.
- If you want to use a shorter cable or you planning to expand the network in future, then star topology is the best choice for you.
- Fully mesh topology is theoretically an ideal choice as every device is connected to every other device.
- If you want to use twisted pair cable for networking, then you should build star topologies.

33. Discuss the basic network architecture.

A set of layers and protocols is called the network architecture. The specification of an architecture must contain enough information to allow an implementer to write the program or build the hardware for each layer so that it will correctly obey the appropriate protocol. Neither the details of the implementation nor the specification of the interfaces is part of the architecture because they are hidden inside the machines and not visible from outside. It is not even necessary that the interfaces on all machines in a network be the same. A basic network architecture showing layers, protocols and interface is given below:

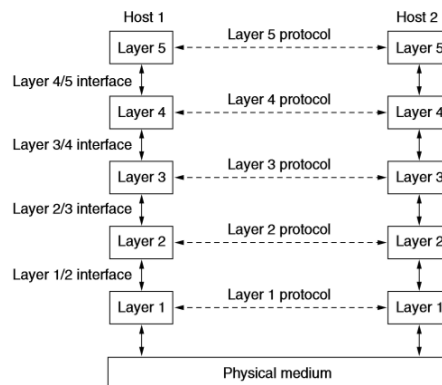


Figure 1-13. Layers, protocols, and interfaces.

34. Write down some of the design issues that should be considered for an efficient network.

The design issues for an efficient network is given below:

I. Addressing:

Every layer needs a mechanism for identifying senders and receivers. That's why addressing is needed in order to specify a specific destination.

II. Data transmission mode:

Data can be transmitted in duplex, half duplex, simplex mode. So rules are required to choose the form of data transfer.

III. Error control:

Physical communication circuits are not perfect. Many error detecting and error correcting codes are known. But both ends of the connection must agree on which one is being issued.

IV. Flow control:

Senders and receivers may not be the same in case of data flow. Senders and receivers can be fast or slow. So data flow must be controlled.

V. Multiplexing:

The process of disassembling, transmitting and assembling is required for long message which should be considered when designing a network.

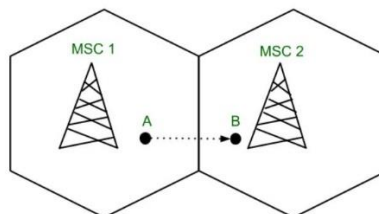
VI. De-multiplexing:

Transmitting data in units that are so small sending each one separately is insufficient. So gathering small messages to transmit as a single large message should be considered designing a network.

35. Describe Handover process

Handover:

In cellular telecommunications, the terms handover or handoff refers to the process of transferring ongoing call or data connectivity from one Base Station to other Base Station. When a mobile moves into the different cell while the conversation is in progress then the MSC (Mobile Switching Center) transfer the call to a new channel belonging to the new Base Station.



When a mobile user A moves from one cell to another cell then BSC 1 signal strength loses for the mobile User A and the signal strength of BSC 2 increases and thus ongoing calls or data connectivity for mobile user goes on without interrupting.

Types of Handoff:

Hard Handoff:

When there is an actual break in the connectivity while switching from one Base Station to another Base Station. There is no burden on the Base Station and MSC because the switching takes place so quickly that it can hardly be noticed by the users. The connection quality is not that good. Hard Handoff adopted the 'break before make' policy.

Soft Handoff:

In Soft Handoff, at least one of the links is kept when radio signals are added or removed to the Base Station. Soft Handoff adopted the 'make before break' policy. Soft Handoff is more costly than Hard Handoff.

36. Discuss Maximum Data Rate of a channel.

As early as 1924, an AT&T engineer, Henry Nyquist, realized that even a perfect channel has a finite transmission capacity. He derived an equation expressing the maximum data rate for a finite-bandwidth noiseless channel. In 1948, Claude Shannon carried Nyquist's work further and extended it to the case of a channel subject to random (that is, thermodynamic) noise (Shannon, 1948). This paper is the most important paper in all of information theory. We will just briefly summarize their now classical results here. Nyquist proved that if an arbitrary signal has been run through a low-pass filter of bandwidth B , the filtered signal can be completely reconstructed by making only $2B$ (exact) samples per second. Sampling the line faster than $2B$ times per second is pointless because the higher-frequency components that such sampling could recover have already been filtered out. If the signal consists of V discrete levels, Nyquist's theorem states:

$$\text{maximum data rate} = 2B \log_2 V \text{ bits/sec}$$

For example, a noiseless 3-kHz channel cannot transmit binary (i.e., two-level) signals at a rate exceeding 6000 bps. So far we have considered only noiseless channels. If random noise is present, the situation deteriorates rapidly. And there is always random (thermal) noise present due to the motion of the molecules in the system. The amount of thermal noise present is measured by the ratio of the signal power to the noise power, called the SNR (Signal-to-Noise Ratio). If we denote the signal power by S and the noise power by N , the signal-to-noise ratio is S/N . Usually, the ratio is expressed on a log scale as the quantity $10 \log_{10} S/N$ because it can vary over a tremendous range. The units of this log scale are called decibels (dB), with "deci" meaning 10 and "bel" chosen to honor Alexander Graham Bell, who invented the telephone. An S/N ratio of 10 is 10 dB, a ratio of 100 is 20 dB, a ratio of 1000 is 30 dB, and so on. The manufacturers of stereo amplifiers often characterize the bandwidth (frequency range) over which their products are linear by giving the 3-dB frequency on each end. These are the points at which the amplification factor has been approximately halved (because $10 \log_{10} 0.5 \approx -3$). Shannon's major result is that the maximum data rate or capacity of a noisy channel whose bandwidth is B Hz and whose signal-to-noise ratio is S/N , is given by:

$$\text{maximum number of bits/sec} = B \log_2 (1 + S/N)$$

37. What is frequency División Multiplexing?

The modulation schemes we have seen let us send one signal to convey bits along a wired or wireless link. However, economies of scale play an important role in how we use networks. It costs essentially the same amount of money to install and maintain a high-bandwidth transmission line as a low-bandwidth line between two different offices (i.e., the costs come from having to dig the trench and not from what kind of cable or fiber goes into it). Consequently, multiplexing schemes have been developed to share lines among many signals. FDM (Frequency Division Multiplexing) takes advantage of passband transmission to share a channel. It divides the spectrum into frequency bands, with each user having exclusive possession of some band in which to send their signal. AM radio broadcasting illustrates FDM. The allocated spectrum is about 1 MHz, roughly 500 to 1500 kHz. Different frequencies are allocated

to different logical channels (stations), each operating in a portion of the spectrum, with the interchannel separation great enough to prevent interference. For a more detailed example, in Fig. 2-25 we show three voice-grade telephone channels multiplexed using FDM. Filters limit the usable bandwidth to about 3100 Hz per voice-grade channel. When many channels are multiplexed together, 4000 Hz is allocated per channel. The excess is called a guard band. It keeps the channels well separated. First the voice channels are raised in frequency, each by a different amount. Then they can be combined because no two channels now occupy the same portion of the spectrum. Notice that even though there are gaps between the channels thanks to the guard bands, there is some overlap between adjacent channels. The overlap is there because real filters do not have ideal sharp edges. This means that a strong spike at the edge of one channel will be felt in the adjacent one as nonthermal noise. This scheme has been used to multiplex calls in the telephone system for many years, but multiplexing in time is now preferred instead. However, FDM continues to be used in telephone networks, as well as cellular, terrestrial wireless, and satellite networks at a higher level of granularity. When sending digital data, it is possible to divide the spectrum efficiently without using guard bands. In OFDM (Orthogonal Frequency Division Multiplexing), the channel bandwidth is divided into many subcarriers that independently send data (e.g., with QAM).

38. What is Time Division Multiplexing?

An alternative to FDM is TDM (Time Division Multiplexing). Here, the users take turns (in a round-robin fashion), each one periodically getting the entire bandwidth for a little burst of time. An example of three streams being multiplexed with TDM is shown in Fig. 2-27. Bits from each input stream are taken in a fixed time slot and output to the aggregate stream. This stream runs at the sum rate of the individual streams. For this to work, the streams must be synchronized in time. Small intervals of guard time analogous to a frequency guard band may be added to accommodate small timing variations.

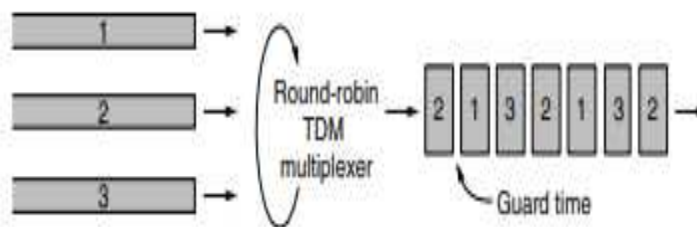


Figure 2-27. Time Division Multiplexing (TDM).

TDM is used widely as part of the telephone and cellular networks. To avoid one point of confusion, let us be clear that it is quite different from the alternative STDM (Statistical Time Division Multiplexing). The prefix “statistical” is added to indicate that the individual streams contribute to the multiplexed stream not on a fixed schedule, but according to the statistics of their demand. STDM is packet switching by another name.

39. What is SONET/SDH?

In the early days of fiber optics, every telephone company had its own proprietary optical TDM system. After AT&T was broken up in 1984, local telephone companies had to connect to multiple long-distance carriers, all with different optical TDM systems, so the need for standardization became obvious. In 1985, Bellcore, the RBOC’s research arm, began working on a standard, called SONET (Synchronous Optical NETwork). Later, ITU joined the effort, which resulted in a SONET standard and a set of parallel ITU recommendations (G.707, G.708, and G.709) in 1989. The ITU recommendations are called SDH (Synchronous Digital Hierarchy) but differ from SONET only in minor ways. Virtually all the long-distance telephone traffic in the United States, and much of it elsewhere, now uses trunks running SONET in the physical layer. For additional information about SONET, see Bellamy (2000), Goralski (2002), and Shepard (2001). The SONET design had four major goals. First and foremost, SONET had to make it possible for different carriers to interwork. Achieving this goal required defining a common signaling standard with respect to wavelength, timing, framing structure, and other issues. Second, some means was needed to unify the U.S., European,

and Japanese digital systems, all of which were based on 64-kbps PCM channels but combined them in different (and incompatible) ways. Third, SONET had to provide a way to multiplex multiple digital channels. At the time SONET was devised, the highest-speed digital carrier actually used widely in the United States was T3, at 44.736 Mbps. T4 was defined, but not used much, and nothing was even defined above T4 speed. Part of SONET's mission was to continue the hierarchy to gigabits/sec and beyond. A standard way to multiplex slower channels into one SONET channel was also needed. Fourth, SONET had to provide support for operations, administration, and maintenance (OAM), which are needed to manage the network. Previous systems did not do this very well. An early decision was to make SONET a traditional TDM system, with the entire bandwidth of the fiber devoted to one channel containing time slots for the various subchannels. As such, SONET is a synchronous system. Each sender and receiver is tied to a common clock. The master clock that controls the system has an accuracy of about 1 part in 10⁹. Bits on a SONET line are sent out at extremely precise intervals, controlled by the master clock. The basic SONET frame is a block of 810 bytes put out every 125 μ sec. Since SONET is synchronous, frames are emitted whether or not there are any useful data to send. Having 8000 frames/sec exactly matches the sampling rate of the PCM channels used in all digital telephony systems. The 810-byte SONET frames are best described as a rectangle of bytes, 90 columns wide by 9 rows high. Thus, $8 \times 810 = 6480$ bits are transmitted 8000 times per second, for a gross data rate of 51.84 Mbps. This layout is the basic SONET channel, called STS-1 (Synchronous Transport Signal-1). All SONET trunks are multiples of STS-1. The first three columns of each frame are reserved for system management information, as illustrated in Fig. 2-39. In this block, the first three rows contain the section overhead; the next six contain the line overhead. The section overhead is generated and checked at the start and end of each section, whereas the line overhead is generated and checked at the start and end of each line. A SONET transmitter sends back-to-back 810-byte frames, without gaps between them, even when there are no data (in which case it sends dummy data). From the receiver's point of view, all it sees is a continuous bit stream, so how does it know where each frame begins? The answer is that the first 2 bytes of each frame contain a fixed pattern that the receiver searches for. If it finds this pattern in the same place in a large number of consecutive frames, it assumes that it is in sync with the sender. In theory, a user could insert this pattern into the payload in a regular way, but in practice it cannot be done due to the multiplexing of multiple users into the same frame and other reasons. The remaining 87 columns of each frame hold $87 \times 9 \times 8 \times 8000 = 50.112$ Mbps of user data. This user data could be voice samples, T1 and other carriers swallowed whole, or packets. SONET is simply a convenient container for transporting bits. The SPE (Synchronous Payload Envelope), which carries the user data does not always begin in row 1, column 4. The SPE can begin anywhere within the frame. A pointer to the first byte is contained in the first row of the line overhead. The first column of the SPE is the path overhead (i.e., the header for the end-to-end path sublayer protocol)

40. Mension Data Link Layer design functions.

The data link layer uses the services of the physical layer to send and receive bits over communication channels. It has a number of functions, including:

- ✓ Providing a well-defined service interface to the network layer.
- ✓ Dealing with transmission errors.
- ✓ Regulating the flow of data so that slow receivers are not swamped by fast senders.

41. Show relationships between packets and frames.

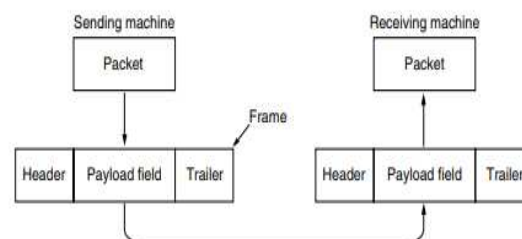


Figure 3-1. Relationship between packets and frames.

42. Describe virtual and actual communications.

The data link layer can be designed to offer various services. The actual services that are offered vary from protocol to protocol. Three reasonable possibilities that we will consider in turn are: 1. Unacknowledged connectionless service. 2. Acknowledged connectionless service. 3. Acknowledged connection-oriented service.

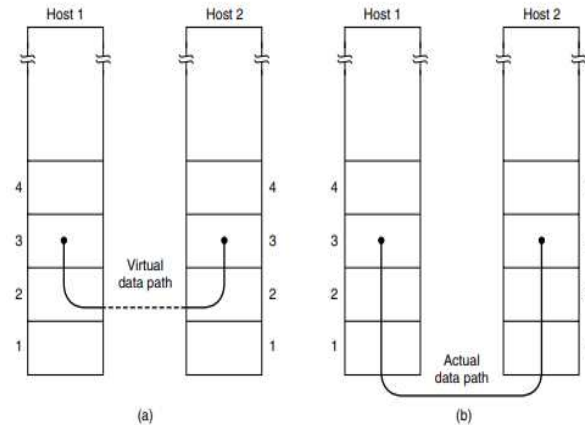


Figure 3-2. (a) Virtual communication. (b) Actual communication.

43. Describe framing and its methods.

Framing: Framing is a point-to-point connection between two computers or devices consists of a wire in which data is transmitted as a stream of bits. However, these bits must be framed into discernible blocks of information. Framing is a function of the data link layer.

We will look at four methods: Byte count, Flag bytes with byte stuffing, Flag bits with bit stuffing, Physical layer coding violations.

44. Describe framing with byte stuffing.

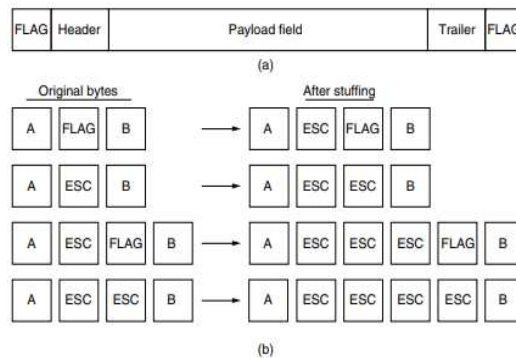


Figure 3-4. (a) A frame delimited by flag bytes. (b) Four examples of byte sequences before and after byte stuffing.

Often the same byte, called a flag byte, is used as both the starting and ending delimiter. The data link layer on the receiving end removes the escape bytes before giving the data to the network layer. This technique is called byte stuffing. Each frame begins and ends with a special bit pattern, 01111110 or 0x7E in hexadecimal. This pattern is a flag byte. Whenever the sender's data link layer encounters five consecutive 1s in the data, it automatically stuffs a 0 bit into the outgoing bit stream. This bit stuffing is analogous to byte stuffing, in which an escape byte is stuffed into the outgoing character stream before a flag byte in the data.

45. Describe error detection and error correction.

Error-Detecting

Whenever a message is transmitted, it may get scrambled by noise or data may get corrupted. To avoid this, we use error-detecting codes which are additional data added to a given digital message to help us detect if an error occurred during transmission of the message. A simple example of error-detecting code is parity check.

Error-Correcting

Along with error-detecting code, we can also pass some data to figure out the original message from the corrupt message that we received. This type of code is called an error-correcting code. Error-correcting codes also deploy the same strategy as error-detecting codes but additionally, such codes also detect the exact location of the corrupt bit.

In error-correcting codes, parity check has a simple way to detect errors along with a sophisticated mechanism to determine the corrupt bit location. Once the corrupt bit is located, its value is reverted (from 0 to 1 or 1 to 0) to get the original message.

46. What are the assumptions of Dynamic channel allocation?

Underlying all the work done in this area are the following five key assumptions:

1. **Independent Traffic.** The model consists of N independent stations (e.g., computers, telephones), each with a program or user that generates frames for transmission. The expected number of frames generated in an interval of length Δt is $\lambda \Delta t$, where λ is a constant (the arrival rate of new frames). Once a frame has been generated, the station is blocked and does nothing until the frame has been successfully transmitted.
2. **Single Channel.** A single channel is available for all communication. All stations can transmit on it and all can receive from it. The stations are assumed to be equally capable, though protocols may assign them different roles (e.g., priorities).
3. **Observable Collisions.** If two frames are transmitted simultaneously, they overlap in time and the resulting signal is garbled. This event is called a collision. All stations can detect that a collision has occurred. A collided frame must be transmitted again later. No errors other than those generated by collisions occur.
4. **Continuous or Slotted Time.** Time may be assumed continuous, in which case frame transmission can begin at any instant. Alternatively, time may be slotted or divided into discrete intervals (called slots). Frame transmissions must then begin at the start of a slot. A slot may contain 0, 1, or more frames, corresponding to an idle slot, a successful transmission, or a collision, respectively.
5. **Carrier Sense or No Carrier Sense.** With the carrier sense assumption, stations can tell if the channel is in use before trying to use it. No station will attempt to use the channel while it is sensed as busy. If there is no carrier sense, stations cannot sense the channel before trying to use it. They just go ahead and transmit. Only later can they determine whether the transmission was successful.

47. Describe CSMA/CD with diagram.

Persistent and nonpersistent CSMA protocols are definitely an improvement over ALOHA because they ensure that no station begins to transmit while the channel is busy. However, if two stations sense the channel to be idle and begin transmitting simultaneously, their signals will still collide. Another improvement is for the stations to quickly detect the collision and abruptly stop transmitting, (rather than finishing them) since they are irretrievably garbled anyway. This strategy saves time and bandwidth. This protocol, known as CSMA/CD (CSMA with Collision Detection), is the basis of the classic Ethernet LAN, so it is worth devoting some time to looking at it in detail. It is important to realize that collision detection is an analog process. The station's hardware must listen to the channel while it is transmitting. If the signal it reads back is different from the signal it is putting out, it knows that a collision is occurring. The implications are that a received signal must not be tiny compared to the transmitted signal (which is difficult for wireless, as received signals may be 1,000,000 times weaker than transmitted signals) and that the modulation must be chosen to allow collisions to be detected (e.g., a collision of two 0-volt signals may well be impossible to detect). CSMA/CD, as well as many other LAN protocols, uses the conceptual model of Fig. 4-5. At the point marked t_0 , a station has finished transmitting its frame. Any other station having a frame to send may now attempt to do so. If two or more stations decide to transmit simultaneously, there will be a collision. If a station detects a collision, it aborts its transmission, waits a random period of time, and then tries again (assuming that no other station has started transmitting in the meantime). Therefore, our model for CSMA/CD will consist of alternating contention and transmission periods, with idle periods occurring when all stations are quiet (e.g., for lack of work).

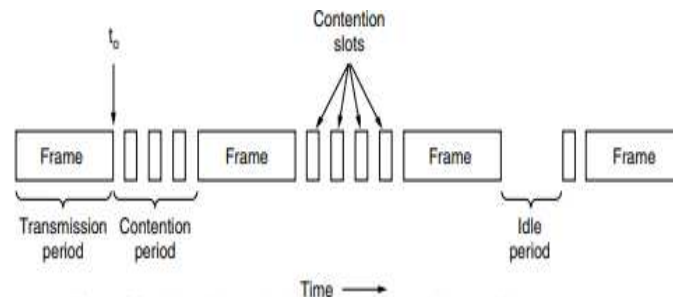


Figure 4-5. CSMA/CD can be in contention, transmission, or idle state.

48. Describe BitMap Protocol with Figure.

In our first collision-free protocol, the basic bit-map method, each contention period consists of exactly N slots. If station 0 has a frame to send, it transmits a 1 bit during the slot 0. No other station is allowed to transmit during this slot. Regardless of what station 0 does, station 1 gets the opportunity to transmit a 1 bit during slot 1, but only if it has a frame queued. In general, station j may announce that it has a frame to send by inserting a 1 bit into slot j . After all N slots have passed by, each station has complete knowledge of which stations wish to transmit. At that point, they begin transmitting frames in numerical order (see Fig. 4-6).

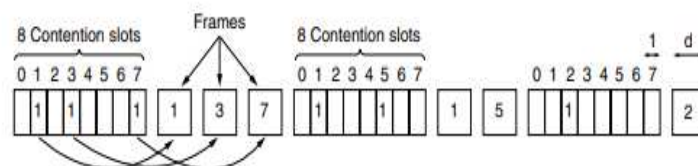


Figure 4-6. The basic bit-map protocol.

Since everyone agrees on who goes next, there will never be any collisions. After the last ready station has transmitted its frame, an event all stations can easily monitor, another N -bit contention period is begun. If a station becomes ready just after its bit slot has passed by, it is out of luck and must remain silent until every station has had a chance and the bit map has come around again.

49. Describe Ethernet and Switch Ethernet with figure.

Ethernet

Ethernet network is used to create local area network and connect multiple computers or other devices such as printers, scanners, and so on. In a wired network, this is done with the help of fiber optic cables, while in a wireless network, it is done through wireless network technology. An Ethernet network uses various topologies such as star, bus, ring, and more.

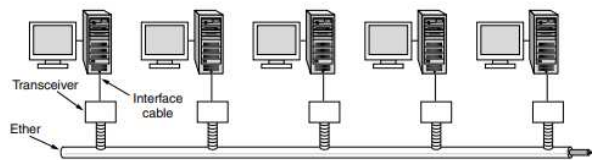


Figure 4-13. Architecture of classic Ethernet.

Switch Ethernet

In switched Ethernet, the hub connecting the stations of the classic Ethernet is replaced by a switch. The switch connects the high-speed backplane bus to all the stations in the LAN. The switch-box contains a number of ports, typically within the range of 4 – 48. A station can be connected in the network by simply plugging a connector to any of the ports. Connections from a backbone Ethernet switch can go to computers, peripherals or other Ethernet switches and Ethernet hubs.

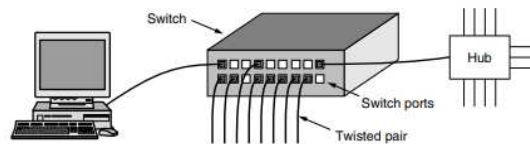


Figure 4-18. An Ethernet switch.

50. What is Hidden Terminal Problem? Describe

The problem of a station not being able to detect a potential competitor for the medium because the competitor is too far away is called the hidden terminal problem.

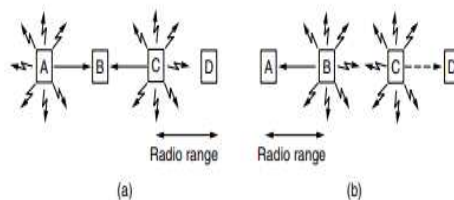


Figure 4-11. A wireless LAN. (a) A and C are hidden terminals when transmitting to B. (b) B and C are exposed terminals when transmitting to A and D.

First consider what happens when A and C transmit to B, as depicted in Fig. 4-11(a). If A sends and then C immediately senses the medium, it will not hear A because A is out of range. Thus C will falsely conclude that it can transmit to B. If C does start transmitting, it will interfere at B, wiping out the frame from A. (We assume here that no CDMA-type scheme is used to provide multiple channels, so collisions garble the signal and destroy both frames.) We want a MAC protocol that will prevent this kind of collision from happening because it wastes bandwidth.

51. What is E-mail? Write the architecture of an Email System.

EMAIL:

Short for electronic mail, e-mail or email is information stored on a computer that is exchanged between two users over telecommunications. More plainly, e-mail is a message that may contain text, files, images, or other attachments sent through a network to a specified individual or group of individuals.

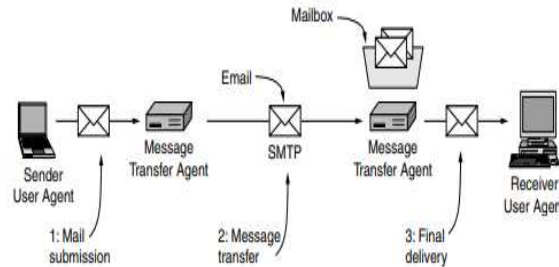


Figure 7-7. Architecture of the email system.

It consists of two kinds of subsystems: the user agents, which allow people to read and send email, and the message transfer agents, which move the messages from the source to the destination. We will also refer to message transfer agents informally as mail servers. The user agent is a program that provides a graphical interface, or sometimes a text- and command-based interface that lets users interact with the email system. It includes a means to compose messages and replies to messages, display incoming messages, and organize messages by filing, searching, and discarding them. The act of sending new messages into the mail system for delivery is called mail submission.

52. Should we compress before encrypting or encrypt after compressing, why? [BUET]

There is no difference in the security provided, but because of the way compression algorithms work, you are probably going to get better compression if you compress first then encrypt.

Compression algorithms exploit statistical redundancies (such as those that exist in natural language or in many file formats) in the data which *should* be eliminated when you encrypt it, therefore an encrypted message *shouldn't* be able to be compressed all that well.

However, lossless data compression algorithms will always fail to compress some files; indeed, any compression algorithm will necessarily fail to compress any data containing no discernible patterns. Attempts to compress data that has been compressed already will therefore usually result in an expansion, as will attempts to compress all but the most trivially encrypted data.

Source: StackOverflow and Wikipedia

53. From this IP address 192.168.10.2/28, Answer the following questions consider IPV4 address –

- a) Subnet Mask
- b) Network address
- c) Broadcast address
- d) First Network address
- e) Last Network Address

Solution:

Subnet Mask

= 11111111.11111111.11111111.11110000

= 255.255.255.240 (Ans – a)

We get change in 4th octet of subnet mask

4th octet of IP = 0 = 00000010

4th octet of subnet mask = 11110000

ANDing 4th octet of IP with subnet mask

we get 00000000 = 0,

So, Network address = 192.168.10.0 (Ans – b)

Here, CIDR = 28

So, making last (32 – 28) = 4 bits into 1 of network address

We get 00001111 = 15

So, broadcast address = 192.168.10.15 (Ans – c)

First address is the next immediate address of Network address

So, First address = 192.168.10.1 (Ans – d)

Last address is the prior immediate address of Broadcast address

So, Last address = 192.168.10.14 (Ans – e)

54. In the following IP address 172.20.0.0/27, find how many subnets and hosts per subnet?

Solution:

Here, given IP address = 172.20.0.0

=172.20.0.00000000

Subnet Mask = 11111111.11111111.11111111.11100000

=255.255.255.224

*(Technique: $11100000 = (2^3 - 1) * 2^5 = 7 * 32 = 224$)*

From 4th octet of subnet mask we get Network bit = 3 and host bit = 5

So, Number of subnet = $2^3 = 8$ (Ans -1)

And hosts per subnet = $2^5 = 32$

And, valid host per subnet = $32 - 2 = 30$ (Ans -2)

55. A network address is given 172.18.10.0/23, divide this network address into 4 subnets and find every subnet address, start address, subnet mask, broadcast address etc.

Solution:

As we should divide the given network address into 4 subnets, we have to find a fixed CIDR for this operation.

Here, total hosts = $2^{(32-23)} = 2^9 = 512$

Now, $512/4 = 128$

We can write $2^7 = 128$

We see, every subnet block will have 128 hosts.

So, our network bit will be $(32-7) = 25$

So, our new CIDR or subnet prefix = /25

1st Subnet:

Subnet address = 172.18.10.0/25

Start address = 172.18.10.1

Subnet mask = 255.255.255.128 (as new CIDR = /25, and $2^7 = 128$)

Broadcast address = 172.18.10.127

2nd Subnet:

Subnet address = 172.18.10.128/25

Start address = 172.18.10.129

Subnet mask = 255.255.255.128 (as new CIDR = /25, and $2^7 = 128$)

Broadcast address = 172.18.10.255

3rd Subnet:

Subnet address = 172.18.11.0/25 (as 4th octet is filled up in previous subnet block, so 3rd octet is changed)

Start address = 172.18.11.1

Subnet mask = 255.255.255.128 (as new CIDR = \25, and $2^7 = 128$)

Broadcast address = 172.18.11.127

4th Subnet:

Subnet address = 172.18.11.128/25

Start address = 172.18.11.129

Subnet mask = 255.255.255.128 (as new CIDR = \25, and $2^7 = 128$)

Broadcast address = 172.18.11.255

56. Given IP address 211.17.180.0/24, Administrator wants to divide it into 32 subnets, so what will be the subnet mask and how many hosts belongs to every subnet, what is the first and last address of 1st and last subnet address?

Solution:

As we should divide the given network address into 32 subnets, we have to find a fixed CIDR for this operation.

Here, total hosts = $2^{(32-24)} = 2^8 = 256$

Now, $256/32 = 8$

We can write $2^3 = 8$

We see, every subnet block will have 8 hosts.

So, our network bit will be $(32-3) = 29$

So, our new CIDR or subnet prefix = /29

So, subnet mask

255.255.255.11111000

=255.255.255.248 (Ans -1)

1st subnet:

Host number per subnet = $2^3 - 2 = 6$ (Ans -2)

First address = 211.17.180.1 (Ans -3)

Last address = 211.17.180.6 (Ans -4)

Last subnet:

Host number per subnet = $2^3 - 2 = 6$ (Ans -5)

First address = 211.17.249 (Ans -6)

Last address = 211.17.180.254 (Ans -7)

(As total host is 256, x.x.x.255 is reserved for broadcast in this case)

57. You have been asked to create a subnet mask for the 172.16.0.0 network. Your organization requires 900 subnets, with at least 50 hosts per subnet. What subnet mask should you use?

Solution:

Given network address is Class B, so 16 bit network address and 16 bit host address.

900 subnets need minimum 10 bits for 3rd and 4th octet. The rest 6 bits (4th octet) is enough for 50 hosts.

So, subnet mask = 255.255.11111111.11000000

= 255.255.255.192 (Ans.)

Read the following topics from the book: Computer Networks by Andrew S. Tanenbaum [5th edition] [pdf book given in the book section]

Page 156 SONET/SDH

Page 320 Bluetooth Architecture

Page 342 Virtual LANS

Page 366 shortest path routing

Page 474 OSPF

Page 644 IMAP

Page 467 ARP

Page 638 SMTP

Page 797 Digital Signature

Page 853 SSL